— Seminar 5 – CD312 – 4.11.2021

quiz ! sp. vect → subsp.

Aplicații liniare

U, V sp. vect, $\dim U = m$; $\dim V = n$; K = corpul scalarilor.

Def: $f: U \to V$ s.n. apl. liniară dc. ⇝ i) $f(u_1+u_2) = f(u_1)+f(u_2)$ $\forall u_1, u_2 \in U$

ii) $f(\lambda u) = \lambda \cdot f(u)$; $\forall u \in U$ $\forall \lambda \in K$.

Obs: f apl lin ⇒ $f(0_U) = 0_V$

$f(0_U) = f(0_U + 0_U) \overset{i)}{=} f(0) + f(0) \Rightarrow f(0_U) = 0_V$. ▨

Ex 1: Studiați liniaritatea:

a) $f: \mathbb{R}^3 \to \mathbb{R}$; $f(x,y,z) = 2xy - z$. Renata

b) $f: \mathbb{R}^3 \to \mathbb{R}$; $f(x,y,z) = 2x + 3y + 7z$. Robert C

c) $f: \mathbb{R}[X] \to \mathbb{R}[X]$; $f(P) = 2P' + 3P''$. Maria ✓

d) $f: \mathbb{R}^4 \to \mathbb{R}^2$; $f(x,y,z,t) = (2x - y, \sin t)$. Robert S

e) $f: M_2(\mathbb{R}) \to \mathbb{R}^3$; $f\begin{pmatrix} a & b \\ c & d \end{pmatrix} = (a-b, d, 2+c)$. Eusebiu

f) $f: \mathbb{R}_{\leq 10}[X] \to \mathbb{R}^3$; $f(P) = (0, P(0) + 2P(1), P'(1))$. Sebastian

Exp 1: $f: \mathbb{R}^2 \to \mathbb{R}$; $f(x,y) = x - y$ e liniară.

i) $u_1 = (x_1, y_1) \in \mathbb{R}^2$, $u_2 = (x_2, y_2) \in \mathbb{R}^2$ $\Rightarrow u_1 + u_2 = (x_1 + x_2, y_1 + y_2)$.

$f(u_1+u_2) = f(\underbrace{x_1+x_2}_{x}, \underbrace{y_1+y_2}_{y}) = (x_1+x_2) - (y_1+y_2) =$

$= (x_1 - y_1) + (x_2 - y_2) = f(x_1, y_1) + f(x_2, y_2) = f(u_1) + f(u_2)$.

ii) $u = (x,y)$; $\lambda \in \mathbb{R} \Rightarrow \lambda u = (\lambda x, \lambda y)$.

$f(\lambda u) = f(\lambda x, \lambda y) = \lambda x - \lambda y = \lambda(x-y) = \lambda f(x,y) = \lambda \cdot f(u)$.

i), ii) ⇒ f liniară.

Exp 2. $f: \mathbb{R}^2 \to \mathbb{R}$; $f(x,y) = x - y + 2$.

$f(0,0) = 2$; Pp. f lin ⇒ $f(0,0) = 0$. ↯ ⇒ f nu e lin.

Exp 3. $f: \mathbb{R}^2 \to \mathbb{R}$; $f(x,y) = x^2 + y^2$. ; $\lambda f(u) = \lambda(x^2+y^2)$.

$u = (x,y)$ $f(0,0) = 0$. ; $f(\lambda u) = f(\lambda x, \lambda y) = \lambda^2 x^2 + \lambda^2 y^2 = \lambda^2(x^2+y^2)$. ⇒ # ⇒ f nu e lin.

f) $f(P) = (0, P(0)+2P(1), P'(1))$.

$P, Q \Rightarrow f(P+Q) = (0, (P+Q)(0)+2\cdot(P+Q)(1), (P+Q)'(1)) =$

$= (0, P(0)+Q(0)+2P(1)+2Q(1), (P'+Q')(1)) =$

$= (0, P(0)+2P(1), P'(1)) + (0, Q(0)+2Q(1), Q'(1))$.

$= f(P) + f(Q)$.

$f(\lambda P) = \lambda f(P)$

Nucleul și imaginea unei aplicații liniare

$f: U \to V$ apl lin; U, V sp vect/K; $\dim U = m$; $\dim V = n$.

$\operatorname{Ker} f = \{u \in U \mid f(u) = 0_V\} \overset{\text{subsp.}}{\subseteq} U \Rightarrow \dim(\operatorname{Ker} f) \leq \dim U$.

$\operatorname{Im} f = \{v \in V \mid \exists u \in U \text{ s.t. } f(u) = v\} \overset{\text{subsp}}{\subseteq} V \Rightarrow \dim(\operatorname{Im} f) \leq \dim V$.

Prop: $\boxed{\dim(\operatorname{Ker} f) + \dim(\operatorname{Im} f) = \dim U}$

Prop f inj $\Leftrightarrow \operatorname{Ker} f = \{0_U\}$.

f surj $\Leftrightarrow \operatorname{Im} f = V$

Exp.4. $f: \mathbb{R}^3 \to \mathbb{R}^3$; $f(x,y,z) = (x-2y, -3z+2y, 3z-x)$.

$\operatorname{Ker} f$; $\operatorname{Im} f$? $\Rightarrow$ o bază, dim. ; ? f izomorfism $\Leftrightarrow$ lin bij.

$\operatorname{Ker} f = \{u \in \mathbb{R}^3 \mid f(u) = 0_{\mathbb{R}^3}\} = \{(x,y,z) \mid f(x,y,z) = (0,0,0)\}$

$= \{(x,y,z) \mid (x-2y, 2y-3z, 3z-x) = (0,0,0)\}$.

$(x,y,z) \in \operatorname{Ker} f \Leftrightarrow \begin{cases} x-2y=0 \\ 2y-3z=0 \\ 3z-x=0 \end{cases} \Rightarrow \boxed{y = \frac{3}{2}z} \Rightarrow \boxed{x = 3z}$

$\operatorname{Ker} f = \{(3z, \frac{3z}{2}, z) \mid z \in \mathbb{R}\} = \{z(3, \frac{3}{2}, 1) \mid z \in \mathbb{R}\}$.

$\operatorname{Ker} f = Sp\{\underbrace{(3, \frac{3}{2}, 1)}_{u_0}\}$; $\{(3, \frac{3}{2}, 1)\}$ l.i. $\Rightarrow$ bază $\Rightarrow \boxed{\dim \operatorname{Ker} f = 1}$

$\{u_0\}$ bază

Pp. f inj $\Rightarrow \operatorname{Ker} f = \{0_U\} \Rightarrow \dim \operatorname{Ker} f = 0$. $\Rightarrow$ ↯ $\Rightarrow$ f nu e inj.

$\dim(\operatorname{Ker} f) + \dim(\operatorname{Im} f) = \dim(\mathbb{R}^3) \Rightarrow \boxed{\dim(\operatorname{Im} f) = 2}$.

$\text{Im } f = \text{Sp}\{f(e_1), f(e_2), f(e_3)\}$. $f(x,y,z) = (x-2y, -3z+2y, 3z-x)$.

$f(e_1) = f(1,0,0) = (1,0,-1) = v_1$ (x y z)

$f(e_2) = f(0,1,0) = (-2,2,0) = v_2 \Rightarrow \{v_1, v_2, v_3\}$ sist de gen pt $\text{Im } f$.

$f(e_3) = f(0,0,1) = (0,-3,3) = v_3$

$\text{Im } f \subseteq \mathbb{R}^3$.

$T = \begin{pmatrix} \overline{v_1} \\ \overline{v_2} \\ \overline{v_3} \end{pmatrix} = \left(\begin{array}{cc|c} 1 & 0 & -1 \\ -2 & 2 & 0 \\ 0 & -3 & 3 \end{array}\right)$; $\det T = 0$.

$\text{rg } T = 2 \Rightarrow v_1, v_2, v_3$ lin dep.

v_1, v_2 lin indep.

$\{v_1, v_2\}$ bază pt $\text{Im } f$; $\dim(\text{Im } f) = 2$.

f nu e surj pt. că $\text{Im } f \neq \mathbb{R}^3$.

$\dim(\text{Im } f) \neq \dim \mathbb{R}^3$. ▨

Matricea atașată unei apl liniare.

$f: U \to V$; $U \to$ bază $B = \{u_1, \dots, u_m\}$. $\dim U = m$.

$V \to$ bază $B' = \{v_1, \dots, v_n\}$; $\dim V = n$

$M_f^{B,B'}$

$f(u_1) = \alpha_{11} \cdot v_1 + \alpha_{12} \cdot v_2 + \dots + \alpha_{1n} \cdot v_n$; $[f(u_1)]_{B'} = \begin{pmatrix} \alpha_{11} \\ \alpha_{12} \\ \vdots \\ \alpha_{1n} \end{pmatrix}$.

$f(u_2) = \alpha_{21} \cdot v_1 + \alpha_{22} \cdot v_2 + \dots + \alpha_{2n} \cdot v_n$

$\dots$

$f(u_m) = \alpha_{m1} \cdot v_1 + \alpha_{m2} \cdot v_2 + \dots + \alpha_{mn} \cdot v_n$.

$M_f^{B,B'} = \left([f(u_1)]_{B'} \mid [f(u_2)]_{B'} \mid \dots \mid [f(u_m)]_{B'}\right) =$

$= \begin{pmatrix} \alpha_{11} & \alpha_{21} & \dots & \alpha_{m1} \\ \alpha_{12} & \alpha_{22} & \dots & \alpha_{m2} \\ \vdots & \vdots & & \vdots \\ \alpha_{1n} & \alpha_{2n} & \dots & \alpha_{mn} \end{pmatrix} : n \times m$.

$\text{rg}(f) = \text{rg}(M_f^{B,B'}) = \dim(\text{Im } f)$.

Exp 5) $f: \mathbb{R}^4 \to \mathbb{R}^4$; $f(x,y,z,t) = (x-y, z+t, z-y, t)$.

$M_f^{B_c} = ?$ $\quad f(x,y,z,t) = x(1,0,0,0) + y(-1,0,-1,0) + z(0,1,1,0) + t(0,1,0,1)$.

$B_c(\mathbb{R}^4) = \{e_1 = (1,0,0,0), e_2, e_3, e_4\}$.

$B_c(\mathbb{R}^4) = \{e_1^t = (1,0,0,0), e_2^t, e_3^t, e_4^t\}$.

$f(e_1) = f(1,0,0,0) = (1,0,0,0)$. $\quad f(e_3) = (0,1,1,0)$

$f(e_2) = f(0,1,0,0) = (-1,0,-1,0)$ $\quad f(e_4) = (0,1,0,1)$.

$$M_f^{B_c} = \left([f(e_1)]_{B_c} \mid [f(e_2)]_{B_c} \mid [f(e_3)]_{B_c} \mid [f(e_4)]_{B_c} \right).$$

$$= \begin{pmatrix} 1 & -1 & 0 & 0 \\ 0 & 0 & 1 & 1 \\ 0 & -1 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} : 4\times 4 \overset{L_2 \leftrightarrow L_3}{\sim} \begin{pmatrix} 1 & -1 & 0 & 0 \\ 0 & -1 & 1 & 0 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 \end{pmatrix}.$$

$rg = 4$.

$rg(M_f^{B_c}) = 4 = \dim(\operatorname{Im} f)$.

$\operatorname{Im} f \subseteq \mathbb{R}^4$.

$\dim(\mathbb{R}^4) = 4$

$\Rightarrow \operatorname{Im} f = \mathbb{R}^4$.

$\Rightarrow f$ surj.

Exp 6) $f: M_2(\mathbb{R}) \to \mathbb{R}_{\leq 2}[X]$.

$f\begin{pmatrix} a & b \\ c & d \end{pmatrix} = dX + cX^2$.

$M_f^{B_c, B_c'} = ?$.

$B_c(M_2(\mathbb{R})) = \{E_{11} = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, E_{12}, E_{21}, E_{22}\}$.

$B_c(\mathbb{R}_{\leq 2}[X]) = \{1, X, X^2\}$.

$f(E_{11}) = f\begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} = 0 = 0 \cdot 1 + 0 \cdot X + 0 \cdot X^2$.

$f(E_{12}) = f\begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} = 0 = 0 \cdot 1 + 0 \cdot X + 0 \cdot X^2$.

$f(E_{21}) = f\begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix} = X^2 = 0 \cdot 1 + 0 \cdot X + 1 \cdot X^2$

$f(E_{22}) = f\begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix} = X = 0 \cdot 1 + 1 \cdot X + 0 \cdot X^2$

$\dim \operatorname{Im} f = 2$; bază pt $\operatorname{Im} f = \{X, X^2\}$.

$$M_f^{B_c, B_c'} = \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix} : 3\times 4.$$

$rg f = rg M_f^{B_c} = 2$.

Ex 7). ? $M_f^{B_c}$; $\ker f$; $\operatorname{Im} f$.

a) $f(x,y,z) = 2x + 3y + 7z$.

b) $f: \mathbb{R}_{\leq 4}[X] \to \mathbb{R}_{\leq 3}[X]$; $f(P) = 2P' + 3P''$.

c) $f: \mathbb{R}_{\leq 3}[X] \to \mathbb{R}^3$; $f(P) = (0, P(0)+2P(1), P'(1))$

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a) $f: \mathbb{R}^3 \to \mathbb{R}$.

$B_c = \{e_1 = (1,0,0), e_2, e_3\}$ → $B_c' = \{1\}$.

$f(e_1) = f(1,0,0) = 2$

$f(e_2) = 3$

$f(e_3) = 7$.

$M_f^{B_c} = (2 \;\; 3 \;\; 7)$: 1×3

$\dim \mathbb{R}^3$ $\qquad$ $\dim \mathbb{R}^3$

$\operatorname{rg} M_f^B = 1 = \dim(\operatorname{Im} f)$. ⇒ $\operatorname{Im} f = \mathbb{R}$ ⇒ f surj.

$\dim_{\mathbb{R}} \mathbb{R} = 1$

bază pt $\operatorname{Im} f = \{1\}$.

$\ker f = \{u \mid f(u) = 0\}$.

$= \{(x,y,z) \mid 2x + 3y + 7z = 0\} = \{(x,y,z) \mid x = -\frac{3}{2}y - \frac{7}{2}z\}$.

$= \{(-\frac{3}{2}y - \frac{7}{2}z, y, z) \mid y, z \in \mathbb{R}\} = \{y(-\frac{3}{2}, 1, 0) + z(-\frac{7}{2}, 0, 1)\}$

v_1 $\qquad$ v_2

$\{v_1, v_2\}$ s. gen pt $\ker f$ ⇒ bază.

l. i.

Verificăm: $\dim(\ker f) = 2$.

$\dim(\ker f) + \dim(\operatorname{Im} f) = \dim \mathbb{R}^3$.

Temă: b), c).